

Energy-momentum tensor and metric near the Schwarzschild sphere

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Introduction

Background

- Quantum effects such as *Hawking radiation* accompany the formation and evolution of black holes.
- Taking these effects into consideration, the question of if, how and when astrophysical black holes develop any of the horizons predicted by *classical general relativity* is still open.
- If the radiation is not terminated when the collapsing object reaches some macroscopic scale or becomes a *Planck-scale* remnant, the notion of an astrophysical black hole being in a permanent state of asymptotic approach to horizon formation becomes untenable.

Results

- Regularity of the horizon radius r_g of a collapsing body constrains a **limiting form of a spherically symmetric energy-momentum tensor $T_{\mu\nu}$** .
- If trapped spacetime regions form in finite time, then the *Einstein equations* imply **violation of the null energy condition**. Here, this is a local result valid even if *future null infinity* is not defined.
- Our findings indicate that the required violation of the *null energy condition* may be **incompatible with the standard analysis of black hole evaporation**.

Assumptions & Methodology

[1] Trapped spacetime regions form in finite time

[2] Spherical symmetry

The most general form of a spherically symmetric metric is given by

$$\begin{aligned} ds^2 &= -e^{2h(t,r)} f(t,r) dt^2 + f(t,r)^{-1} dr^2 + r^2 d\Omega & (\text{Schwarzschild coordinates}) \\ &= -e^{2h_u(u,r)} f_u(u,r) du^2 - 2e^{h_u(u,r)} dudr + r^2 d\Omega & (\text{retarded null coordinates}) \\ &= -e^{2h_v(v,r)} f_v(v,r) dv^2 + 2e^{h_v(v,r)} dvdr + r^2 d\Omega & (\text{advanced null coordinates}) \end{aligned}$$

[3] Regularity of the horizon radius

The *trace* $T := T^\mu_\mu \equiv R/8\pi$ and the *square* $\mathfrak{T} := T^{\mu\nu}T_{\mu\nu}$ of the energy-momentum tensor $T_{\mu\nu}$ are assumed to have arbitrary but finite values (“*no drama at the horizon*” *postulate*):

$$|T(z, r_g)| < \infty \quad \text{and} \quad |\mathfrak{T}(z, r_g)| < \infty$$

[4] Semiclassical gravity

In the *semiclassical Einstein field equations*, the curvature terms are related to the expectation value of the renormalised energy-momentum tensor, i.e.

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi\langle\hat{T}_{\mu\nu}\rangle \equiv 8\pi T_{\mu\nu}$$

Limiting form of the (tr) block of $T_{\mu\nu}$

Regularity of the horizon radius r_g of a collapsing body constrains a limiting form of a spherically symmetric energy-momentum tensor near it.

$$T_b^a = \begin{pmatrix} -\Xi/f & -se^{-h\Xi}/f^2 \\ se^{h\Xi} & \Xi/f \end{pmatrix}, \quad T_{ab} = \frac{\Xi}{f} \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix} \quad \text{where} \quad s = \pm 1$$

This result is independent of the Einstein equations and remains valid even if higher-order curvature terms are added to their left hand side. The four cases of the limiting form of $T_{\mu\nu}$ can be summarised as follows.

$\text{sgn}(T_{tt})$	$\text{sgn}(T_t^r)$	s	Vaidya metric		Einstein equations have real solutions	$\Xi = -\Upsilon^2 < 0$
–	–	+	(v, r)	$M'_t(v) < 0$		
–	+	–	(u, r)	$M'_u(u) > 0$	Einstein equations do not have real solutions	$\Xi = -\Upsilon^2 > 0$
+	–	–	(u, r)	$M'_u(u) < 0$		
+	+	+	(v, r)	$M'_t(v) > 0$		

- ⇒ Close to the Schwarzschild sphere, geometry can always be described by either an ingoing or outgoing *Vaidya metric* with increasing or decreasing mass function, **but if trapped regions form in finite time, there are only two options.**

Violation of the null energy condition (NEC)

For $\Xi = -\Upsilon^2 < 0$ and $s = \pm 1$ the null energy condition is violated:

$$T_{\hat{a}\hat{b}}k^{\hat{a}}k^{\hat{b}} < 0 \quad \text{for a radial null vector} \quad k^{\hat{a}} = (1, s, 0, 0)$$

Implications

- For classical matter that satisfies the NEC, trapped regions cannot form in finite time.
- If quantum effects are strong enough to violate the NEC, the event horizon (if it exists) is located inside the apparent horizon.
- The finite-time Schwarzschild radius r_g is timelike during both the expansion and contraction of the trapped region.

References

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Thin shell formalism

A brief overview:

- Toy model for the gravitational collapse of a massive, spherically symmetric thin dust shell (hypersurface Σ) sandwiched between flat interior (i.e. *Minkowski*) and curved exterior (e.g. *Vaidya*) geometry.
- Shell dynamics is determined by so-called **junction conditions** $\equiv$ rules for joining both sides of Σ prescribed by the *Einstein field equations (EFE)*.
- The joined geometry is a solution of the *EFE* with an additional distributional $T_{\mu\nu}$ concentrated on Σ .

1st junction condition

The induced metric h_{ab} is the same on both sides of the shell:

$$ds_\Sigma^2 = h_{ab}dy^a dy^b = -d\tau^2 + R^2 d\Omega_{D-1}$$

2nd junction condition

The surface energy-momentum density is related to the discontinuity of the extrinsic curvature K across the two sides $\Sigma^\pm$ of the hypersurface,

$$S_{ab} = -([K_{ab}] - [K]h_{ab})/8\pi,$$

where $K := K_a^a$ and $[K] := K|_{\Sigma^+} - K|_{\Sigma^-}$. The **equation of motion** for the shell is

$$\mathcal{D}(R) := \frac{\sqrt{F + \dot{R}^2} - \sqrt{1 + \dot{R}^2}}{R} - \frac{\ddot{R}}{\sqrt{1 + \dot{R}^2}} + \frac{2\ddot{R} + F'}{2\sqrt{F + \dot{R}^2}} = 0$$

→ Interesting quantity to monitor: gap $X := R - r_g$

Thin shell dynamics with evaporation

Evaporating shell and positive energy density

$$\mathcal{D}(R) + \frac{F_U}{F\sqrt{F + \dot{R}^2}} \left(\frac{1}{2} - \dot{R}\dot{U} \right) = 0$$

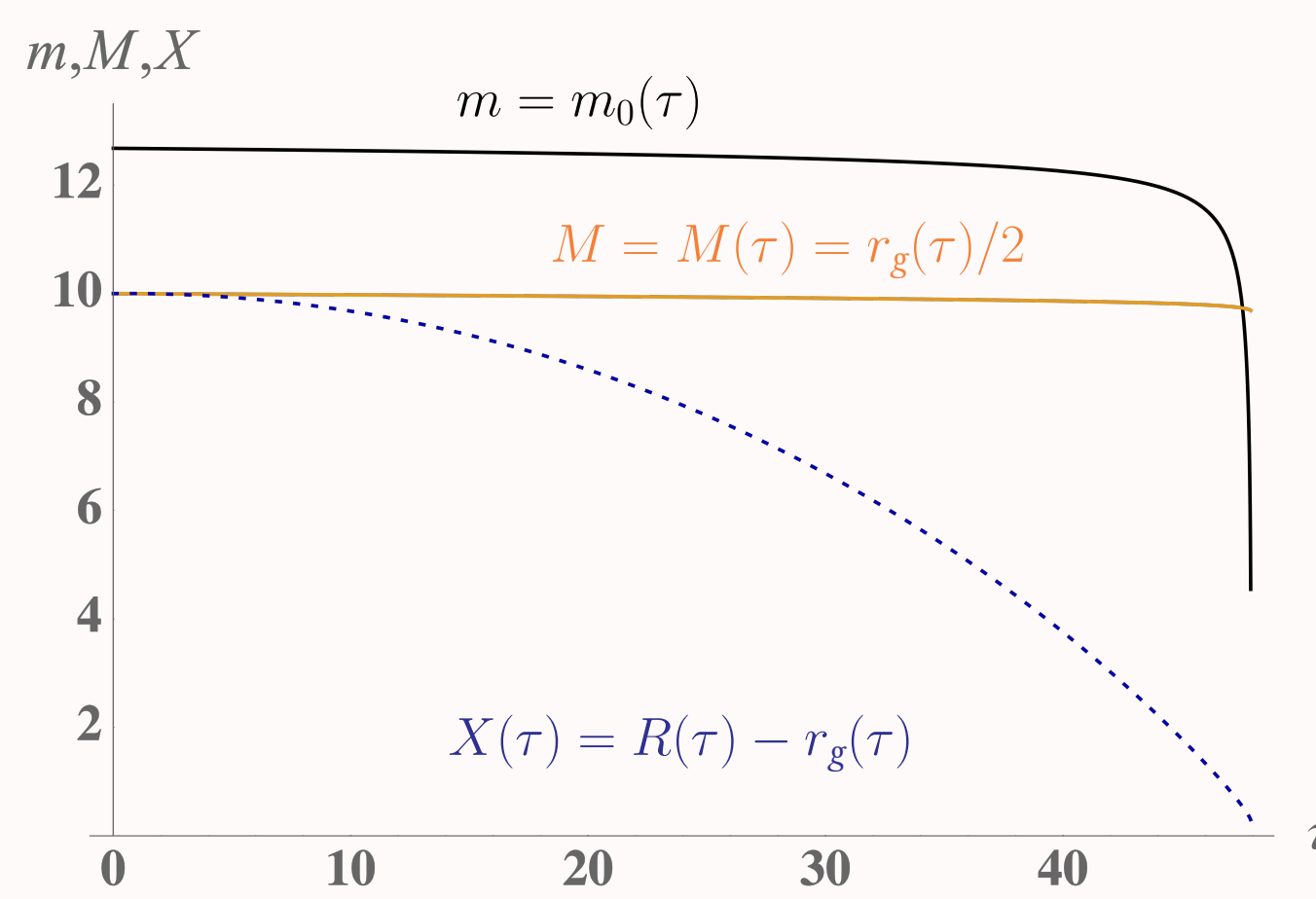


Figure 1: Transition to null trajectory.

Evaporating shell and negative energy density

$$\mathcal{D}(R) - \frac{F_V}{F\sqrt{F + \dot{R}^2}} \left(\frac{1}{2} + \dot{R}\dot{V} \right) = 0$$

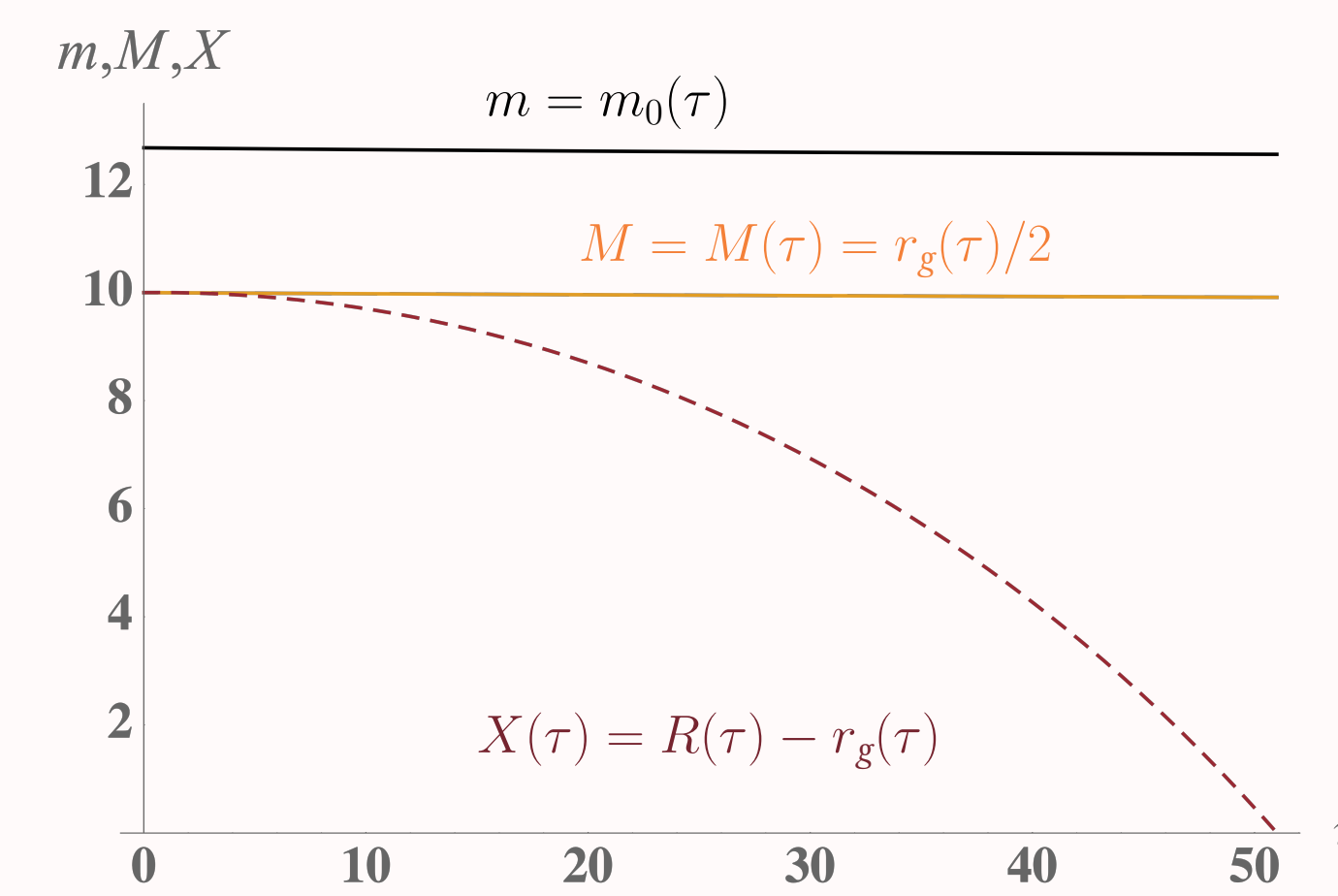


Figure 2: Horizon crossing.

Comparison with the classical trajectory

Initial conditions for simulation

$$R(0) = 0, \quad \dot{R} = 0, \quad \alpha = 1 \\ r_0 = r_g(0) = 20$$

Results

- Collapse is accelerated, shell becomes null at $\tau = 47.912$.
- Collapse is delayed.

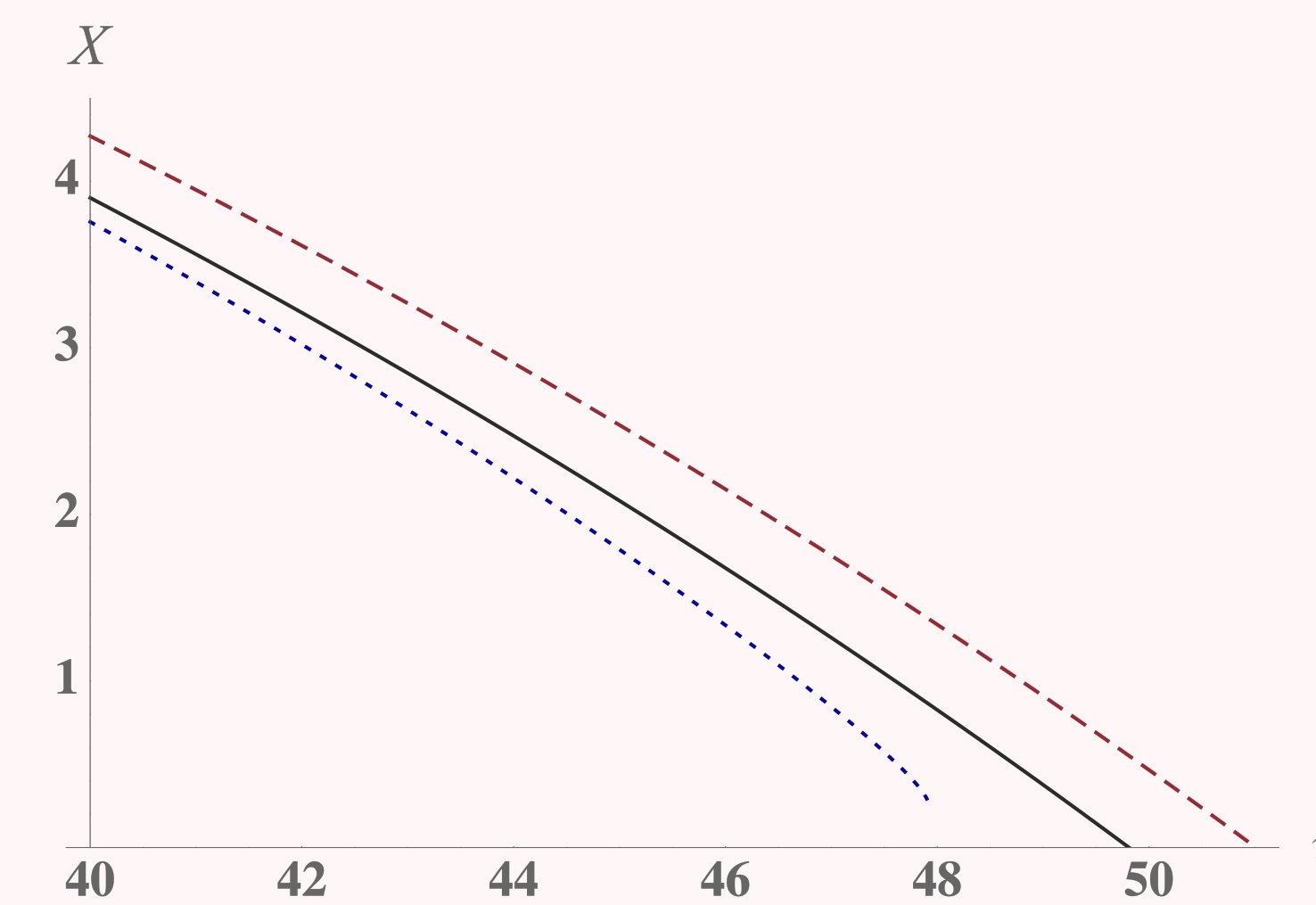


Figure 3: Comparison of the classical trajectory (grey line) with the evaporating shell with exterior geometry given by the **outgoing Vaidya metric** with $dr_g/du = -\alpha/r_g^2$ and the ingoing Vaidya metric with $dr_g/dv = -\alpha/r_g^2$.

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